

Forge Agent OS - Master Specification

Forge - Complete Vision Document

A next generation Agent Operating System built on a VS Code foundation. The goal is to combine the usability of modern AI editors with the learning capabilities inspired by Hermes style agent systems.

Project Background

The concept originated from analysing VS Code based approaches, Void, Zed and Hermes Agent. The objective is not to clone any of them, but to create a new branded product with its own identity.

Reference Technologies

Void Repository: <https://github.com/voideditor/void> Zed Repository: <https://github.com/zed-industries/zed> Hermes Agent Repository: <https://github.com/nousresearch/hermes-agent> These projects are references for ideas, architecture patterns and inspiration only.

Why VS Code Instead of Zed

Zed offers exceptional performance but is significantly harder to customize. VS Code provides a mature extension ecosystem, React based UI opportunities and faster product iteration.

Product Vision

Forge becomes a workspace where coding, automation, business building, research, content creation, branding and operations are executed by coordinated AI agents.

Core Principles

Local first. Multi agent. Human approval. Persistent learning. Fast UX. Visual workflows. Tool driven actions. Provider agnostic model routing.

Agent Architecture

Architect, Planner, Researcher, Coder, QA, Reviewer, Designer, Marketing, Automation and Operations agents. Each agent may use a different model and provider.

Model Routing

Coder may use local Qwen or DeepSeek. QA may use Claude. Research may use Gemini or OpenRouter. Users decide the model assigned to every role.

Brain System

Brain replaces the concept of simple memory. It stores Project Memory, Skill Memory, User Preferences, Mission History and Architectural Decisions.

Skill Learning

Inspired by Hermes. When an agent solves a difficult problem, a reusable skill is generated and indexed for future use.

Mission System

Users launch missions rather than chats. Example: Create a plumber website. Build a CRM. Automate invoice processing. Research competitors.

Kanban System

Planning, Research, Working, Review and Completed columns. Every mission is visible. Every agent action is traceable.

Workspace Modes

Development Mode, Automation Mode, Marketing Mode, Business Mode, Design Mode and Research Mode.

Blueprint Generation

Before implementation the system creates architecture.md, requirements.md, design.md, brand.md, landing.md, seo.md and execution plans.

Diff Engine

Every code modification must be visible. Users can inspect, approve or reject changes. Similar to professional code review workflows.

Automation Engine

Agents can create files, execute scripts, operate terminals, run Docker containers, browse websites and automate workflows.

Web Search Layer

Web search is a platform service available to all agents. Local models gain access to current information through retrieval and search pipelines.

Tool Registry

Filesystem, Terminal, Git, Docker, SSH, Browser Automation, Web Search, MCP, Image Generation, PDF Generation, Spreadsheet Generation, Email Drafting, Database Tools.

Permissions

Read only actions require no approval. File modifications, terminal commands, deployments and destructive operations require escalating approval levels.

User Experience

Chat remains available in a sidebar for follow ups. The primary experience is mission driven, visual and operational.

Live Feed

Users see exactly what agents are doing: searching, planning, editing files, running tests, generating assets or executing automations.

Marketing Workflows

Generate brand systems, landing pages, ad campaigns, customer personas, SEO strategies and content calendars.

Design MD System

Design documents become reusable assets. Brand identity, colors, typography, layout rules and UX decisions are stored as project knowledge.

MCP Ecosystem

Support for MCP servers allows integrations with databases, business systems, cloud infrastructure and custom enterprise tools.

Deployment

Single installer experience. User downloads the application, adds API keys or connects local models and immediately starts working.

Future Roadmap

Marketplace, team collaboration, shared brains, cloud synchronization, enterprise governance and advanced autonomous workflows.

Competitive Positioning

Cursor focuses on coding. Zed focuses on performance. Hermes focuses on autonomous learning. Forge combines visual workflows, memory, tools, automation and multi agent orchestration.

Success Metric

The product should feel like hiring a team rather than opening an editor.